LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang
RP: **Muhammad Faris Bin Abd Manap**

SEMESTER: OCT 2026 – FEB 2027 (SESI I 2026/2027)

COLLABORATIVE TEACHING (CGCT) — MODEL 1: COLLABORATION IN CONTENT DEVELOPMENT

Collaborative group	**MEQ545 Engineering Economics — CGCT Model 1 (Collaboration in Content Development), Sesi I 2026/2027**
Collaborative members	**Muhammad Faris Bin Abd Manap [336994] — Responsible Person (RP) / content lead; Muhd Azhar Bin Zainol [336907] — teaching team**
Industry panel	**En Ahmad Khairul Nizam Bin Bahrinal — Branch Manager Principal, Homefield Real Estate Sdn Bhd**
Developed content	**MEQ545 All-Quiz Tutorial Portal (web application) — 10 tutorial sets / 107 questions. Live: https://meq545-all-quiz-portal-hub.pages.dev/**
Attribution	**Blue = developed by the teaching team (RP-led); Red = co-developed with the Industry Panel**

Note: This collaboration is additive. The official syllabus, Course Outcomes (CO1–CO3), assessment structure and weights remain unchanged.

Course Outline

Course	**Bachelor of Manufacturing Engineering Technology**
Subject	**Engineering Economics (MEQ545)**
Contact Hours	**3 Credit Hours**

Course Outcomes
Upon completion of this course, students should be able to:

CO1 Explain the basic principles and practice of project management. [PO1, LO1]- {SK1, SK2, SP1, SP2, SP3} (C2)
CO2 Display the principles of the engineering economy to manage process design. [PO10, LO10] (A5)
CO3 Recognise the need for and independently propose engineering economic related analysis in solving economic problems. [PO11, LO9] - {SK8} (C4)

Course Description
This course covers topics that include the principles, fundamental concepts, and methodologies of engineering economics. It emphasizes engineering economic analysis and facilitates rational decision-making regarding costs in environmental engineering practices. Additionally, the course introduces basic concepts and principles of project management, with a focus on planning, scheduling, monitoring, controlling, evaluating, and terminating projects.

Assessment: 100% Coursework

Individual Assignment	: 20%
Group Project	: 20%
Test 1	: 30%
Test 2	: 30%
Total	**: 100%**

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

Recommended Textbook:

William G. Sullivan, Elin M. Wicks, C. Patrick Koelling, Engineering Economy, 20th Ed, Prentice Hall, **2024**, ISBN: 9780133439271

References:

1. Park, C.S, Fundamentals of Engineering Economics, 4th Ed, Pearson, 2019, ISBN: n/a
2. Leland T Blank, Anthony Tarquin, Engineering Economy, 8th Ed, McGraw-Hill Education, 2017, ISBN: 9780073523439
3. Timothy J Kloppenborg, Contemporary Project Management, 4th Ed, Cengage, 2018, ISBN: 9788131518717

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

SYLLABUS
ENGINEERING ECONOMICS – MEQ545

FACULTY OF MECHANICAL ENGINEERING
UNIVERSITI TEKNOLOGI MARA PULAU PINANG BRANCH

CHAPTER	CONTENT/SUBCHAPTER
Chapter 1 Introduction to Engineering Economy (6 hours)	1.1 Engineering Economy 1.2 The Principles of Engineering Economy 1.3 Engineering Economy and The Design Process 1.4 Developing Investment Alternative
Chapter 2: Cost Concepts and Design Economics (6 hours)	2.1 Cost Terminology 2.2 The General Economic Environment 2.3 Cost-Driven Design Optimization
Chapter 3: Cost Estimation Techniques (6 hours)	3.1 Introduction 3.2 An Integrated Approach 3.3 Selected Estimating Techniques (Models) 3.4 Parametric Cost Estimating
Chapter 4: The Time Value of Money (6 hours)	4.1 Money has A Time Value 4.2 Simple Interest 4.3 Compound Interest 4.3.1 Developing Model Repayment 4.3.2 Developing Net Cash-Flow Table 4.4 Finding Interest 4.5 Summary of Interest Formulas and Relationships for Discrete Compounding/Continuous Compounding
Chapter 5: Evaluating a Single Project (6 hours)	5.1 Introduction of Evaluating Capital Project 5.2 Determining the Minimum Attractive Rate of Return (MARR) 5.3 The Present Worth Method 5.4 Bond Value 5.5 The Future Worth Method 5.6 The Internal Rate of Return Method 5.7 The External Rate of Return Method
Chapter 6: Comparison and Selection among Alternatives (4 hours)	6.1 Introduction: Feasibility of Design Alternatives 6.2 Basic Concepts for Comparing Alternatives 6.3 Useful Lives Are Equal & Inequal to the Study Period 6.4 Personal Finances
Chapter 7: Depreciation and Income Taxes (4 hours	7.1 Introduction of Depreciation 7.2 Depreciation Concepts and Terminology 7.3 The Classical (Historical) Depreciation Methods 7.3.1 Straight Line (SL) Method 7.3.2 Declining Balance (DB) Method 7.3.3 DB Switching SL 7.4 Basic Concept of Income Taxes 7.5 Corporate Taxes in Malaysia

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

TEACHING PLAN SESI I 2026/2027 (OCT2026 – FEB2027)

Week	Date	Syllabus	Activity(s)	Developed Content (CGCT Model 1)
1	28/09/2026 - 04/10/2026	**INTRO ENG ECO**	-	—
2	05/10/2026 - 11/10/2026	**CHAPTER 1**	Tutorial 1	**T1–T2 sets (Ch. 1): teaching team (RP-led); deployed & completed**
3	12/10/2026 - 18/10/2026	**CHAPTER 1**	Tutorial 2	T2 set (Ch. 1): completion & feedback; e-attendance captured
4	19/10/2026 - 25/10/2026	**CHAPTER 2**	Tutorial 3	**T3–T4 sets (Ch. 2): teaching team (RP-led); scheduled release**
5	26/10/2026 - 01/11/2026	**CHAPTER 2**	Individual Assignment	—
6	02/11/2026 - 08/11/2026	**CHAPTER 3**	Individual Assignment	—
7	09/11/2026 - 15/11/2026	**CHAPTER 3**	**TEST 1**	—
8	16/11/2026 - 22/11/2026	**CHAPTER 4**	Tutorial 4	**T5 set (Ch. 3 & 4): industry co-developed; scheduled release**
9	23/11/2026 - 29/11/2026	**CHAPTER 4**	Group Project (Report)	—
10	30/11/2026 - 06/12/2026	**CHAPTER 5**	Group Project (Report)	—
11	07/12/2026 - 13/12/2026	**CHAPTER 5**	Tutorial 5	**T6–T7 sets (Ch. 5): industry co-developed; scheduled release**

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

12	14/12/2026 - 20/12/2026	**CHAPTER 6**	Tutorial 6	**T8 set (Ch. 6): industry co-developed; scheduled release**
-	21/12/2026 - 27/12/2026	-	Mid Semester Break	—
13	28/12/2026 - 03/01/2027	**CHAPTER 6**	Tutorial 7	**T9–T10 sets (Ch. 6): team extension of industry framework**
14	04/01/2027 - 10/01/2027	**CHAPTER 7**	TEST 2	—
	11/01/2027 - 17/01/2027		Rev Week	—
	18/01/2027 - 24/01/2027		Submission Group Project	—
	18/01/2027 - 07/02/2027		**NO FINAL EXAMINATION**	—

CGCT MODEL 1 — COLLABORATION WORKFLOW (CONTENT DEVELOPMENT)

1. Topic mapping — the RP maps each tutorial set to the course syllabus (Chapters 1–7) and the Course Outcomes (CO1–CO3).

2. Content development — the RP develops the question sets, worked solutions and answer explanations in the MEQ545 All-Quiz Tutorial Portal; the teaching team reviews and contributes items (sets T1–T4).

3. Industry co-development — the Industry Panel co-develops and validates the industry-context scenarios for sets T5–T8: cost estimation, time value of money, project evaluation and comparison of alternatives (feasibility, cost optimisation, operation & maintenance decisions).

4. Team extension — the teaching team extends the industry-co-developed framework into additional sets T9–T10.

5. Deployment & monitoring — sets are released in the portal according to the schedule above; submissions, scores and attendance e-signatures are monitored via the portal admin dashboard.

Attribution legend: blue = developed by the teaching team (RP-led); red = co-developed with the Industry Panel. Developed content total: 10 tutorial sets / 107 questions (78 multiple-choice, 3 true/false, 24 short-answer, 2 long-answer). Portal link: https://meq545-all-quiz-portal-hub.pages.dev/

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

Tutorial

Tutorial	**Topic**
1	Introduction to Engineering Economy
2	Cost Concepts and Design Economics
3	Cost Estimation Techniques
4	The Time Value of Money
5	Evaluating a Single Project
6	Comparison and Selection among Alternatives
7	Depreciation and Income Taxes

Assessment:

No	**Assessment**	**Marks**	**CO1-PO1**	**CO2-PO10**	**CO3-PO11**
1	Individual Assignment	20%	/		
2	Group Project	20%		/	
3	Test 1	30%		/	
4	Test 2	30%			/

1. Individual Assignment

Topic	Cognitive Level	Context
Chapter 1 (10%)	2 Questions (C1)	**Introduction to Engineering Economy**
Chapter 2 (10%)	2 Questions (C2)	**Cost Concepts and Design Economics**

2. Group Project

Topic	Cognitive Level	Context
Chapter 2 (5%)	Case Study (C3)	**Cost Concepts and Design Economics**
Chapter 3 (5%)		**Cost Estimation Techniques**
Chapter 4 (10%)		**The Time Value of Money**

3. Test 1

Topic	Cognitive Level	Context
Chapter 1 (10%)	1 Questions (C1)	**Introduction to Engineering Economy**
Chapter 2 (10%)	2 Questions (C2)	**Cost Concepts and Design Economics**
Chapter 3 (10%)	1 Questions (C4)	**Cost Estimation Techniques**

4. Test 2

LESSON PLAN MEQ 545 (ENGINEERING ECONOMICS)

Bachelor of Manufacturing Engineering Technology with Honours
Mechanical Engineering Studies, Faculty of Mechanical Engineering
Universiti Teknologi MARA, Cawangan Pulau Pinang

Topic	Cognitive Level	Context
Chapter 4 (10%)	1 Questions (C3)	**The Time Value of Money**
Chapter 5 (10%)	1 Questions (C4)	**Evaluating a Single Project**
Chapter 6 (5%)	1 Questions (C5)	**Comparison and Selection among Alternatives**
Chapter 7 (5%)	1 Questions (C5)	**Depreciation and Income Taxes**

*Assessment subject to change without prior notice